Troubleshooting Guide and Q&A

Access SpheroQuaNNt using https://nlealdovaizem.shinyapps.io/SpheroQuaNNt-v2/

1. The application takes some time to load

This is expected. SpheroQuaNNt uses a U-Net-based image analysis model that must be loaded when the application starts. Depending on the current server status and available resources, the initial loading process may take up to approximately one minute. When using the application for the first time, please allow the loading process to finish before uploading an image.

> **Tip:** If the application appears to be loading, please wait rather than refreshing the page immediately.

The application may also need to restart after a period without activity because shinyapps.io can stop an application instance after it has been idle. The current documentation indicates an **Instance Idle Timeout of 15 minutes by default**.

2. What type of images can I upload?

SpheroQuaNNt is specifically designed to analyze images of spheroids. The application uses the image to estimate the spheroid's dimensions, particularly its **major and minor diameters**, which are subsequently used for volume estimation.

Therefore:

Recommended input

Upload an image containing a **clearly visible spheroid**, preferably:

- A single spheroid or clearly identifiable spheroid;
- Good image contrast;
- Minimal background interference;
- The spheroid clearly visible within the image;
- An image with sufficient resolution for identifying its boundaries.

Do not use

The application is **not designed for arbitrary biological images**, including:

- Individual cells;
- Tissue sections;
- Microscopy fields containing many unrelated objects;
- Images of tissues or animals;
- Graphs or plots;

Important: The application may technically allow you to upload an image that is not a spheroid. However, this does **not** mean that the result will be valid. Additionally, if more than one spheroid appears in the image, only the largest one will be used for volume estimation.

3. What happens if I upload an image that is not a spheroid?

The application does not necessarily prevent the upload of an inappropriate image. If an image does not correspond to the type of object used to train the U-Net model, the model may attempt to segment structures that do not represent a spheroid.

This can produce:

Incorrect segmentation → incorrect radius estimation → incorrect volume estimation.

In some cases, an inappropriate image may also cause the application to fail or disconnect.

Therefore, the numerical result should only be interpreted when the uploaded image represents a spheroid, and the detected structure is biologically plausible. The application does not validate whether the uploaded image is a spheroid. Users are responsible for ensuring that the input image is appropriate for the intended analysis.

4. The application shows an error or disconnects

If the application suddenly stops responding, displays a grey screen, or shows a message such as: **"Disconnected from server"**

Try the following:

Step 1 — Refresh the page

Refresh the browser page and wait for the application to load again.

Step 2 — Wait for initialization

Because the U-Net model needs to be initialized, the application may take some time to become available after restarting.

Step 3 — Try the analysis again

Upload the spheroid image again and repeat the analysis.

Step 4 — If the problem persists

Try:

1. Closing the browser tab;
2. Opening the application again;
3. Using a different browser;
4. Checking that the uploaded file is a valid image;

5. Trying a smaller or higher-quality image.

Step 5 — If the problem persists

Send the problematic file with a brief description of this occurrence to Nathalia Leal Dovaizem (nathalialeal@usp.br) or Alexis Murillo Carrasco (agmurilloc@gmail.com)

5. Why can the application fail even though my computer has enough RAM?

This is an important distinction. The analysis may work perfectly on a local computer with, for example:

16 GB RAM + Intel i7 processor

but the online application does not use the RAM or CPU of your computer to run the U-Net model.

The computation occurs on the server hosting the Shiny application.

For shinyapps.io, the default large instance provides 1 GB of RAM, and the available resources depend on the hosting plan. Posit also notes that shinyapps.io does not guarantee a particular number or speed of CPU cores.

Consequently:

Local performance and online performance are not necessarily equivalent.

A model that requires substantial memory during initialization or inference may therefore behave differently online. If you wish to work SpheroQuaNNt locally, please download reference files from https://github.com/nleald/SpheroQuaNNt

6. The application worked before, but now it does not

There are several possible explanations.

A. The application instance has gone to sleep

shinyapps.io automatically stops instances after they have been idle for a specified period. When someone accesses the application again, a new instance is started. The first load can therefore take longer.

Solution: wait for the application to initialize.

B. The application has restarted

shinyapps.io can shut down and recreate application instances, including for maintenance or recovery from server problems.

Solution: refresh the application and wait for initialization.

C. The application has exceeded its available resources

If the application exceeds the memory available to its instance, the process may be terminated. Posit specifically recommends checking the application logs when "Disconnected from server" occurs.

D. The free-plan usage limit has been reached

The Free shinyapps.io plan has historically included **25 active hours per month**. Posit's documentation defines active hours based on how long application instances are running; importantly, the idle period before sleep also contributes to active-hour consumption.

Therefore, if the application is widely used, it is possible for the hosting account to reach its monthly allowance.

7. The application is temporarily unavailable

If the application does not open at all, this does not necessarily indicate a problem with the image or with your computer.

Possible causes include:

- Application restart;
- Server-side resource limitations;
- Temporary infrastructure problems;
- Application initialization failure;
- Free-plan usage limitations.

Try again after a short period.

If the application displays **"Disconnected from server"**, refreshing the page is the first recommended troubleshooting step. Posit recommends consulting the application logs to identify the underlying error when these disconnections occur.

8. Important interpretation warning

SpheroQuaNNt provides an automated estimate, not a direct physical measurement.

The reliability of the final volume estimate depends on:

1. The quality of the input image;
2. Whether the image actually contains a spheroid;
3. The ability of the U-Net model to correctly segment the spheroid;
4. The accuracy of the estimated major and minor diameters;
5. The assumptions used for calculating spheroid volume.

Therefore, users should visually inspect the segmentation and the resulting measurements before interpreting the estimated volume. An output generated by the application should not automatically be considered valid simply because the application produces a numerical value.